Annotated Bibliography

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References

[1] Irene Aldridge. *High-Frequency Trading: A Practical Guide to Algorithmic Strategies and Trading Systems*. John Wiley & Sons, 2nd edition, 2013.

Aldridge provides a practical overview of high-frequency trading systems, including strategy design, infrastructure requirements, and execution mechanics. The book discusses latency optimization, order routing, and the interaction between predictive signals and real-world exchange behavior. This source is central to my project because it directly addresses the system-level concerns of high-frequency trading. My project focuses on architectural tradeoffs between prediction accuracy and execution latency, and Aldridge's discussion of infrastructure constraints and execution risk informs that design. Although written for practitioners, the book is widely cited and provides technical depth. It supports my emphasis on treating latency and execution rather than abstracting them away, aligning closely with the goals of my simulation framework.

[2] Jean-Philippe Bouchaud, J. Doyne Farmer, and Fabrizio Lillo. How markets slowly digest changes in supply and demand. *Handbook of Financial Markets: Dynamics and Evolution*, pages 57–160, 2009.

[3] Thomas Fischer and Christopher Krauss. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669, 2018.

This paper explores how Long Short-Term Memory (LSTM) neural networks can be used to predict stock market movements. The authors compare LSTM models with traditional machine learning methods, such as random forests and logistic regression, using historical stock data. Their results show that LSTMs perform better at identifying long-term patterns in financial time-series data. This source is useful from a computer science perspective because it clearly explains how deep learning models are structured, trained, and evaluated. It also discusses challenges such as overfitting and changing market behavior. The paper is peer-reviewed and widely cited, making it a credible source. It provides a strong foundation for studying machine learning approaches to stock

prediction and can be extended by incorporating economic or policy-related data.

[4] Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.

This paper studies how modern machine learning methods can be used to predict stock returns using a large number of financial and economic variables. The authors test several models, including neural networks and tree-based methods, and find that machine learning approaches perform better than traditional linear models. Although the paper is published in a finance journal, its focus is strongly related to computer science because it emphasizes model comparison, feature selection, and evaluation methods. The authors are well-known researchers, and the paper is peer-reviewed, which adds to its credibility. This source is relevant to my Independent Study because it shows how economic and policy-related data can be included in machine learning models and highlights the computational challenges of working with large datasets.

[5] Lawrence Harris. *Trading and Exchanges: Market Microstructure for Practitioners*. Oxford University Press, 2003.

Harris offers a comprehensive introduction to market microstructure, explaining how exchanges operate, how orders are matched, and how information is incorporated into prices. The book details the mechanics of limit order books, liquidity provision, and trading costs. This source is foundational for understanding how real markets function. My trading system simulation relies on accurately modeling order matching, queue position, and slippage, all of which are explained in this text. Although written for practitioners, it is academically respected and frequently cited in finance research. It strengthens the realism of my execution simulator and ensures that architectural experiments are grounded in actual exchange behavior rather than unrealistic assumptions.

[6] Michail Patsiarikas, George Papageorgiou, and Christos Tjortjis. Using machine learning on macroeconomic, technical, and sentiment indicators for stock market forecasting. *Information*, 16(2), 2025.

This paper examines stock market forecasting using machine learning models that combine macroeconomic indicators, technical market data, and sentiment information. The authors compare several models, including Random Forests, gradient boosting, and neural networks, to predict stock index movements. By including macroeconomic data, the study connects broader economic and policy conditions to market behavior. From a computer science perspective, this paper is helpful because it focuses on data preprocessing, feature engineering, and comparing different machine learning models. The study is peer-reviewed and clearly explains its methods, making it a reliable source. For my Independent Study, this paper is especially relevant because it shows how economic and policy signals can be integrated into predictive models for stock markets.

[7] Michael Stonebraker, Uğur Çetintemel, and Stan Zdonik. The 8 requirements of real-time stream processing. *Association for Computing Machinery Special Interest Group on Management of Data Record*, 34(4):42–47, 2005.

Stonebraker, Çetintemel, and Zdonik outline the core requirements of real-time stream processing systems, including low latency, determinism, and scalability. The paper discusses architectural decisions that influence throughput and tail latency in event-driven environments. This source is critical for the computer science foundation of my project. My trading system is implemented as an event-driven architecture processing high-frequency data streams. The paper provides guidance on how architectural choices affect latency distributions, which directly connects to my research questions about prediction effectiveness under time constraints. As a publication in a major ACM venue, it is credible and technically rigorous. It supports the systems-engineering perspective of my project.

[8] Álvaro Cartea, Sebastian Jaimungal, and José Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015.

Cartea, Jaimungal, and Penalva present a rigorous treatment of algorithmic and high-frequency trading, combining stochastic modeling with market microstructure theory. The book analyzes limit order books, optimal execution, and the dynamics of order flow. This source is highly relevant to my project because it provides the theoretical foundation for modeling realistic execution behavior. Concepts such as price–time priority, adverse selection, and queue dynamics directly inform the design of my limit order book simulator. Published by Cambridge University Press and authored by leading researchers, it is a credible academic reference. It supports the research questions in my project by linking microstructure theory with the practical challenges of implementing prediction-aware execution systems under latency constraints.